Reductions and NP-completeness

Exercise sheet 9

Exercise 9.1. Let $A \leq_p B$ and $B \leq_p C$.

(a) Prove that $A \leq_p C$.

(b) If $C \in \mathrm{P}$, what can we say about A?

(c) If A is NP-complete, what can we say about C?

Exercise 9.2. Prove that CLIQUE is NP-complete.

Exercise 9.3. An independent set in a graph is a set of vertices with no edges between them. Prove that the problem INDEPENDENT-SET, which asks whether G has an independent set of size at least k, is NP-complete.

Exercise 9.4. A vertex cover is a set of vertices that contains at least one endpoint of every edge. Prove that the problem VERTEX-COVER, which asks whether G has a vertex cover of size at most k, is NP-complete.

Exercise 9.5. * A nondeterministic Turing machine may have several possible transitions for a state and observed symbol. It accepts an input if some computation halts in the accepting state, and rejects if all computations reject. It runs in polynomial time if every computation halts within a polynomial number of steps.

Prove that $L \in \mathrm{NP}$ if and only if a polynomial-time nondeterministic Turing machine decides L.

Exercise 9.6. * Let NEXP be the class of languages decided by exponential-time nondeterministic Turing machines. Prove that if $\mathrm{P} = \mathrm{NP}$, then $\mathrm{EXP} = \mathrm{NEXP}$.